

1) eigenvectors and eigenvalues for:

$$B = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$$

$$B\vec{x} = \lambda\vec{x} \rightarrow B\vec{x} - \lambda\vec{x} = 0 \rightarrow (B - \lambda I)\vec{x} = 0$$

this will have non-trivial solution if $|(B - \lambda I)| = 0$

eigenvalues can be computed: 1.) Find determinant of $(B - \lambda I)$. Determinant will be Quadratic

2.) Solutions to this will give me eigenvalues

$$|B - \lambda I| = \left| \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = \left| \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = \left| \begin{bmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{bmatrix} \right| = (2-\lambda)(2-\lambda) - (3 \cdot 3) = 4 - 2\lambda - 2\lambda + \lambda^2 - 9 = \lambda^2 - 4\lambda - 5.$$

det. of 2x2 matrix = a.d - b.c

Now find solutions of this
Solutions = eigenvalues

$$\lambda^2 - 4\lambda - 5 = 0$$

discriminant

$$D = b^2 - 4ac$$

$$D = 16 - (4 \cdot 1 \cdot -5)$$

$$D = 36$$

Now 2

$$\lambda_{1,2} = \frac{-b \pm \sqrt{D}}{2a} = \frac{4 \pm 6}{2}$$

$$\lambda_1 = 5$$

$$\lambda_2 = -1$$

eigenvectors: can be computed $(B - \lambda I) \cdot \vec{n} = 0$. Now I have everything so calculate $\vec{x}$.

1.) $\vec{n}$ for λ_1

2.) $\vec{n}$ for λ_2

a) eigenvector for $\lambda_1 = -1 \rightarrow (B + 1I) \cdot \vec{n} = 0$

$$\left(\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \right) \cdot \vec{n} = 0$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \cdot \vec{n} = 0$$

$$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{matrix} r_1 \leftarrow r_1 + -3 \cdot r_2 \\ r_2 \leftarrow r_2 - \frac{1}{3} \cdot r_1 \end{matrix}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{matrix} n_1 + n_2 = 0 \\ n_1 = -n_2 \end{matrix}$$

$$\boxed{n_1 = -n_2} \rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \vec{n}_2$$

$\hookrightarrow$ if $n_1 = 1$

b) eigenvector for $\lambda_2 = 5 \rightarrow (B - 5I) \cdot \vec{x} = 0$

$$\left(\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \right) \cdot \vec{x} = 0$$

$$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \cdot \vec{x} = 0$$

$$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{matrix} r_1 \leftarrow r_1 + r_2 \\ r_2 \leftarrow r_2 - r_1 \end{matrix}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{matrix} n_1 - n_2 = 0 \\ n_1 = n_2 \end{matrix} \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \vec{n}_1$$

$\hookrightarrow$ if $n_1 = 1$

Bases of entire eigenspace of $B = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

Diagonalise matrix using $Q \rightarrow$ orthonormal matrix $\rightarrow Q^T \cdot B \cdot Q = D$

1) calculate eigenvalues $\rightarrow \lambda_1 = 5, \lambda_2 = -1$

2) calculate eigenvectors $\rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

3) normalize eigenvectors $\rightarrow$ to make sure that Q will remain orthonormal and $Q^T \cdot Q = I$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow 1^2 + (-1)^2 = c^2$$

$$c = \sqrt{2}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{\text{normalize}} 1^2 + 1^2 = c^2$$

$$c = \sqrt{2}$$

normalized

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \xrightarrow{\text{simplify}} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}$$

$\vec{n}_2$ normalized

normalized

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \xrightarrow{\text{simplify}} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$\vec{n}_1$ normalized

4) orthonormal matrix Q . cds of Q are normalized ei vectors.

$$Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

Since Q is orth., so $Q \cdot Q^T = Q^T \cdot Q = I$

$$\Downarrow$$

$$Q = Q^T$$

5) $\Lambda \rightarrow$ Diag. matrix $\rightarrow$ cds are eigenvalues IN ORDER AS eigenvectors in Q

$$\Lambda = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

6.) Verify that Q orthogonally diagonalizes B by:

$$Q \wedge Q^T = B$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{5\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{5\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} =$$

$$\begin{bmatrix} \left(\frac{5\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}\right) & \left(\frac{5\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2} \cdot -\frac{\sqrt{2}}{2}\right) \\ \left(\frac{5\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2}\right) & \left(\frac{5\sqrt{2}}{2} \cdot -\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2} \cdot -\frac{\sqrt{2}}{2}\right) \end{bmatrix} = \begin{bmatrix} 5/2 - 1/2 & 5/2 + 1/2 \\ 5/2 + 1/2 & 5/2 - 1/2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} = B$$

conclusion: Q orthogonally diagonalizes B because $Q \wedge Q^T = B$

$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ as linear combination of v_1 and v_2

$$a \cdot v_1 + b \cdot v_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 2 \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{bmatrix} \begin{array}{l} R_1 \leftarrow R_1 \cdot \frac{2}{\sqrt{2}} \text{ (so } \frac{2}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} = 1) \\ R_2 \leftarrow -1 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 4/\sqrt{2} \\ 1 & -1 & 0 \end{bmatrix} \begin{array}{l} R_1 \leftarrow R_1 + R_2 \end{array}$$

$$\begin{bmatrix} 2 & 0 & 4/\sqrt{2} \\ 1 & -1 & 0 \end{bmatrix} \begin{array}{l} R_1 \leftarrow R_1 \cdot \frac{1}{2} \end{array}$$

$$\begin{bmatrix} 1 & 0 & 2/\sqrt{2} \\ 1 & -1 & 0 \end{bmatrix} \rightarrow a = 2/\sqrt{2} \rightarrow a = b \rightarrow b = \frac{2}{\sqrt{2}}$$

$\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{2}{\sqrt{2}} \cdot v_1 + \frac{2}{\sqrt{2}} \cdot v_2 \rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ as lin. combination of v_1 and v_2

closed formula:

$$B^n \begin{bmatrix} 2 \\ 0 \end{bmatrix} = B^n \left(\frac{2}{\sqrt{2}} \cdot v_1 + \frac{2}{\sqrt{2}} \cdot v_2 \right)$$

$$B^n \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{2}{\sqrt{2}} \cdot B^n \cdot v_1 + \frac{2}{\sqrt{2}} \cdot B^n \cdot v_2$$

if v_1, v_2 are eigenvectors of B , with corresponding eigenvalues λ_1, λ_2 , then $B^n \cdot v_1 = \lambda_1^n \cdot v_1$
 $B^n \cdot v_2 = \lambda_2^n \cdot v_2$

$$B^n \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{2}{\sqrt{2}} \cdot \lambda_1^n \cdot v_1 + \frac{2}{\sqrt{2}} \cdot \lambda_2^n \cdot v_2 \rightarrow \text{closed formula for } B^n \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

② Page rank is defined if $\rightarrow$ 1.) Existence of eigenvector for eigenvalue $\lambda=1$. (if column stochastic)
 $\rightarrow$ 2.) Uniqueness $\rightarrow$ unique importance vector $\rightarrow$ unique ranking of pages

Prob. of ending on page at the end of rand. walk

$$\begin{aligned} x_0 &= \frac{1}{2} x_2 \\ x_1 &= \frac{1}{2} x_0 + x_3 \\ x_2 &= \frac{1}{2} x_0 + x_1 \\ x_3 &= \frac{1}{2} x_2 \end{aligned}$$

$$\begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1 \\ 1/2 & 1 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\vec{x} = A \cdot \vec{x}$
 (transition probability matrix) $\rightarrow$ eigenvector

is A column STOCHASTIC?

1/2 $\rightarrow$ Probabilities ≥ 0 ✓

2/2 $\rightarrow$ Probabilities in Columns sums to 1. ✓

2/2 conditions satisfied $\Rightarrow$ COLUMN STOCHASTIC

$\downarrow$
 it has eigenvector for eigenvalue $\lambda=1$

$$\begin{aligned} x_0 &= \frac{1}{2} x_2 \\ x_1 &= \frac{1}{2} x_0 + x_3 \\ x_2 &= \frac{1}{2} x_0 + x_1 \\ x_3 &= \frac{1}{2} x_2 \end{aligned}$$

$\rightarrow$ I have x_2 values in x_0 and x_3 .
 therefor I will substitute x_0, x_3 in x_1 and x_2

$$x_0 = \frac{1}{2} x_2$$

$$x_1 = \frac{1}{2} \left(\frac{1}{2} x_2 \right) + \frac{1}{2} x_2$$

$$x_2 = \frac{1}{2} \left(\frac{1}{2} x_2 \right) + \frac{1}{2} \left(\frac{1}{2} x_2 \right) + \frac{1}{2} x_2$$

$$x_3 = \frac{1}{2} x_2$$

$$x_0 = \frac{1}{2} x_2$$

$$x_1 = \left(\frac{1}{4} + \frac{1}{2} \right) x_2$$

$$x_2 = \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{2} \right) x_2$$

$$x_3 = \frac{1}{2} x_2$$

$$x_0 = \frac{1}{2} x_2$$

$$x_1 = \frac{3}{4} x_2$$

$$x_2 = x_2$$

$$x_3 = \frac{1}{2} x_2$$

So importance of pages is:

1.) Page 2

2.) Page 1

3. and 4.) Page 0 and Page 3
↳ equally important

Importance score of website/page is rate of nodes that are pointing to page.

OUTPUT OF PYTHON → RANDOM SURFER
→ PAGERANK

running file: p2p-Gnutella08-mod.txt, with steps: 100

First 10 most visited nodes in random surfer: dict_keys([266, 250, 38, 1901, 128, 179, 249, 145, 753, 754])

Approx number of iterations to get stable state in pagerank: 7

First 10 most visited nodes in page rank: {249: 2.2367140797305805, 127: 2.21553495317447, 266: 2.207001482905122, 367: 2.19854576577225, 123: 2.1610069844687163, 264: 2.1019634349917222, 145: 2.0822777588763866, 427: 2.064981721797204, 352: 2.063130967511946, 251: 2.059634907690215}